

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and data scientist with experience in statistical model development, independent validation, hypothesis testing, and machine learning across financial risk, insurance pricing, and large-scale dependent data. Builds reproducible Python workflows on AWS and HPC, evaluates model assumptions and performance, and translates technical findings into clear business and model-risk recommendations.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Platforms: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL, R, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, cached artifacts, and restartable analytical pipelines

Data Science and Model Risk: Statistical model development and validation, hypothesis testing, MLE/GLS, Monte Carlo simulation, regression, GLM/GBM, machine learning, time-series and spatiotemporal modeling, model selection, diagnostics, feature attribution, and performance evaluation

INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and independently validated a Sequential Probability Ratio Test for binary risk indicators, translating a monitoring problem into explicit hypotheses, likelihood-ratio boundaries, and controlled decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate boundary-crossing probabilities, stopping behavior, Type I/II error, and early-decision tradeoffs; documented model assumptions, performance, and limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Evaluated model performance and feature-attribution patterns and presented findings as pricing and risk-segmentation recommendations for business stakeholders.

RESEARCH AND ENGINEERING

Ph.D. Dissertation Research - Scalable Gaussian Process Inference and Diagnostics

Sep 2024 - Present

- Large-Scale Model Development:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with ordered conditional models to fit up to 145,008 observations per day.
- Data Quality and Processing:** Built observation masks, quality filters, spatial-distance thresholds, hour-specific coordinate offsets, and nearest-neighbor matching to convert incomplete, irregular satellite observations into analysis-ready grids.
- Model Validation and Root-Cause Analysis:** Designed scale- and frequency-resolved diagnostics to distinguish systematic acquisition artifacts and measurement noise from covariance-model misspecification and to identify where model assumptions fail.
- Research Engineering:** Implemented reusable Python/PyTorch workflows with GLS profiling, L-BFGS optimization, CPU/GPU execution, cached artifacts, and restartable HPC jobs for estimation, simulation, and diagnostics.

Longitudinal Observational Study - EITC and Household Labor Supply

2021

- Estimated the relationship between Earned Income Tax Credit exposure and household labor supply using fixed-effects regression and propensity-score matching with household income and demographic covariates; evaluated comparison-group construction, confounding assumptions, and interpretation of model results.

ADDITIONAL

Coursework: Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)